

Laplace Transforms – Partial Table

$f(t) = \mathcal{L}^{-1}[F(s)]$	$F(s) = \mathcal{L}[f(t)]$
c	$\frac{c}{s}$
e^{at}	$\frac{1}{s-a}$
t^n (for $n = 1, 2, 3, \dots$)	$\frac{n!}{s^{n+1}}$
$\sin(at)$	$\frac{a}{s^2 + a^2}$
$\cos(at)$	$\frac{s}{s^2 + a^2}$
$e^{at} \sin(bt)$	$\frac{b}{(s-a)^2 + b^2}$
$e^{at} \cos(bt)$	$\frac{s-a}{(s-a)^2 + b^2}$
$\frac{dy}{dt}$	$s \cdot \mathcal{L}[y] - y(0)$
$\frac{d^2y}{dt^2}$	$s^2 \cdot \mathcal{L}[y] - s \cdot y(0) - y'(0)$
$u_a(t)$	$\frac{e^{-as}}{s}$
$u_a(t) \cdot f(t-a)$	$e^{-as} \cdot \mathcal{L}[f(t)]$